MADHU CHERUKURI

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Career Objective

Pre-final year Computer Science student aspiring for SDE roles in AI development and backend engineering, with hands-on experience in building ML-based applications and API integrations. Experienced through research internships and industry programs, with strong skills in Python, Generative AI, and backend development. Interested in applying AI solutions to real-world problems.

TECHNICAL SKILLS

Languages: Python (Advanced), Java, JavaScript, SQL | **Frameworks:** Spring Boot, React.js, JDBC
AI/ML: Machine Learning, Generative AI, Prompt Engineering, LLM APIs
Backend & APIs: REST API Design, OSRM APIs, Stripe Payment Gateway | **Databases:** MySQL
Tools & Platforms: Git, GitHub, LeetCode (DSA), HTML5, CSS, VS Code | **GenAI Tools:** ChatGPT, Midjourney, Runway, ElevenLabs

EXPERIENCE

Backend Engineering Intern | **Infosys Springboard — Cohort 6.0** | *Feb 2026 – Mar 2026*

- Engineered a full-stack Smart Ride-Sharing & Carpooling Platform using React, Spring Boot, and MySQL, supporting multi-role workflows (passenger, driver, admin) end-to-end.
- Integrated OSRM APIs for real-time dynamic distance calculation and route optimization
- Implemented Stripe Payment Gateway for secure transactions and built automated ride confirmation notification pipeline reducing driver response latency.

AI/ML Research Intern | **IIITDM Kancheepuram — Prof. Sk. Noor Mahammad** | *Sep 2025 – Nov 2025*

- Designed and deployed a Hybrid ML model for real-time ransomware detection, analyzing file operations, process activity, and behavioral signatures from system-level datasets.
- Applied ensemble ML techniques to classify malicious vs. benign process behavior, achieving measurable improvement in detection precision over baseline models.
- Curated and preprocessed multi-feature system activity datasets, enabling high-fidelity behavioral pattern recognition for cybersecurity threat intelligence.

Backend Development Intern | **VaultOfCodes (AICTE)** | A u g – *2025*

- Built a Library Management System backend in Java with JDBC-MySQL integration, implementing CRUD operations for 1,000+ book and user records with efficient relational data persistence.

PROJECTS

Kisan AI Agricultural Assistant Platform — *· OpenRouter LLM API · OpenWeatherMap API · data.gov.in API · REST APIs*

- Built a scalable AI agricultural assistant in a 4-member team by integrating LLM, weather, and government data APIs, focusing on backend API integration and reliable fallback architecture for rural farmers.

AI-Powered Emergency Triage Management System— *XGBoost · Scikit-learn · Flask-SocketIO · SQLite · Pandas*

- Developed an AI-powered hospital triage system that reduced emergency patient prioritization time from 20 minutes to 3 seconds using XGBoost and real-time dashboards.

AI-Powered Commercial Advertisement — *ChatGPT · Midjourney · Runway · ElevenLabs*

- Produced an end-to-end AI-generated advertisement leveraging LLM scriptwriting, diffusion-based visual generation, AI animation, and voice.

EDUCATION

B.Tech — Computer Science Engineering | Anurag Engineering College, Kodad | *2023–2027* | *GPA: 8.0/10*

Senior Secondary — MPC | KRR Govt Junior College, Kodad | *2021–2023* | *CGPA: 73.9/10*

ACHIEVEMENTS & CERTIFICATIONS

Certifications: Generative AI (Google Cloud) • Python AI Development • Infosys Springboard (Multiple) • Internshala Student Partner
Hackathons: Adobe India Hackathon • SRM Smart Park Hackathon • National Level Tech Fest (Coding Contest & Logo Design)

Additional Information:: **Reference**